

Math 550, S20
Exam 2

Name: *Answer*

Instructions:

- There are totally 5 problems.
- You must show all of your work to receive a credit for a correct answer.
- No calculator is allowed in the exam.

Problem	Points	Score
1	<i>20</i>	
2	<i>22</i>	
3	22	
4	20	
5	<i>20</i>	
Total	<i>104</i>	

Good Luck !

Problem 1. (20 points) Find the local maximum and minimum values and saddle points of the function $f(x, y) = x^2y - 6y^2 - 3x^2$.

$$f(x, y) = x^2y - 6y^2 - 3x^2$$

$$\Rightarrow Df(x, y) = (2xy - 6x, x^2 - 12y)$$

$$\begin{cases} f_x = 0 \\ f_y = 0 \end{cases} \Rightarrow \begin{cases} 2xy - 6x = 0 \\ x^2 - 12y = 0 \end{cases} \Rightarrow \begin{cases} 2x(y - 3) = 0 \\ y = \frac{x^2}{12} \end{cases} \Rightarrow$$

$$2x\left(\frac{x^2}{12} - 3\right) = 0 \Rightarrow x(x^2 - 36) = 0$$

$$\Rightarrow x(x+6)(x-6) = 0$$

$$\Rightarrow x = 0, -6, 6$$

$$\begin{matrix} \downarrow & \downarrow & \downarrow \\ y = 0 & y = 3 & y = 3 \end{matrix}$$

$$\Rightarrow \text{critical points} = (0, 0), (-6, 3), (6, 3)$$

$$\text{Next, } Hf(x, y) = \begin{bmatrix} f_{xx} & f_{xy} \\ f_{xy} & f_{yy} \end{bmatrix} = \begin{bmatrix} 2y - 6 & 2x \\ 2x & -12 \end{bmatrix}$$

$$\text{For } (0, 0), Hf(0, 0) = \begin{bmatrix} -6 & 0 \\ 0 & -12 \end{bmatrix} \Rightarrow \begin{matrix} d_1 = -6 < 0 \\ d_2 = -12 < 0 \end{matrix}$$

$\Rightarrow (0, 0)$ gives a local maximum value

$$f(0, 0) = 0$$

$$\text{For } (-6, 3), Hf(-6, 3) = \begin{bmatrix} 0 & -12 \\ -12 & -12 \end{bmatrix} \Rightarrow \begin{matrix} d_1 = 0 \\ d_2 = -144 < 0 \end{matrix}$$

$\Rightarrow (-6, 3)$ is a saddle point.

$$\text{For } (6, 3), Hf(6, 3) = \begin{bmatrix} 0 & 12 \\ 12 & -12 \end{bmatrix} \Rightarrow \begin{matrix} d_1 = 0 \\ d_2 = -144 < 0 \end{matrix}$$

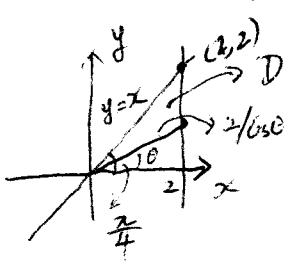
$\Rightarrow (6, 3)$ is a saddle point.

Problem 2. (22 points)

(a) Evaluate $\int_{\frac{1}{2}}^1 \int_0^{2y} \cos(\pi y^2) dx dy$.

$$\begin{aligned} \int_{1/2}^1 \int_0^{2y} \cos(\pi y^2) dx dy &= \int_{1/2}^1 \cos(\pi y^2) [x]_{x=0}^{x=2y} dy \\ &= \int_{1/2}^1 2y \cos(\pi y^2) dy \\ &= \left[\frac{1}{\pi} \sin(\pi y^2) \right]_{y=1/2}^{y=1} \\ &= \frac{1}{\pi} (\sin \pi - \sin \frac{\pi}{4}) \\ &= \frac{1}{\pi} (0 - \frac{\sqrt{2}}{2}) = -\frac{\sqrt{2}}{2\pi} \end{aligned}$$

(b) Let D be the region enclosed by the lines $y = x$, $y = 0$, and $x = 2$. Evaluate the double integral $\iint_D \frac{x^2}{(x^2 + y^2)^{3/2}} dy dx$ using **polar coordinates**.



$$D = \{(x, y) \mid 0 \leq x \leq 2, 0 \leq y \leq x\}$$

$$\downarrow x = r \cos \theta, y = r \sin \theta$$

$$D^* = \{(r, \theta) \mid 0 \leq \theta \leq \frac{\pi}{4}, 0 \leq r \leq \frac{2}{\cos \theta}\}$$

Thus

$$\begin{aligned} \iint_D \frac{x^2}{(x^2 + y^2)^{3/2}} dy dx &= \int_0^{\pi/4} \int_0^{2/\cos \theta} \frac{(r \cos \theta)^2}{(r^2)^{3/2}} r dr d\theta \\ &= \int_0^{\pi/4} \int_0^{2/\cos \theta} \cos^2 \theta dr d\theta \\ &= \int_0^{\pi/4} \cos^2 \theta [r]_{r=0}^{r=2/\cos \theta} d\theta \\ &= \int_0^{\pi/4} \cos^2 \theta \cdot \left(\frac{2}{\cos \theta}\right) d\theta \\ &= \int_0^{\pi/4} 2 \cos \theta d\theta \\ &= 2 \sin \theta \Big|_{\theta=0}^{\theta=\pi/4} = 2(\sin \frac{\pi}{4} - 0) \\ &= 2 \cdot \frac{\sqrt{2}}{2} = \sqrt{2} \end{aligned}$$

Problem 3. (22 points)

(a) Evaluate $\int_1^2 \int_3^x \int_0^{\sqrt{3}y} \frac{z}{y^2 + z^2} dz dy dx$.

$$\begin{aligned}
 \int_1^2 \int_3^x \int_0^{\sqrt{3}y} \frac{z}{y^2 + z^2} dz dy dx &= \int_1^2 \int_3^x \left[\frac{1}{2} \ln(y^2 + z^2) \right]_{z=0}^{z=\sqrt{3}y} dy dx \\
 &= \int_1^2 \int_3^x \frac{1}{2} (\ln(4y^2) - \ln(y^2)) dy dx \rightarrow \ln(4y^2) - \ln(y^2) = \ln \frac{4y^2}{y^2} = \ln 4 \\
 &= \int_1^2 \int_3^x \frac{1}{2} \ln 4 dy dx \\
 &= \int_1^2 (\ln 2) (x-3) dx \\
 &= (\ln 2) \left(\frac{x^2}{2} - 3x \right) \Big|_{x=1}^{x=2} \\
 &= (\ln 2) \left[(2-6) - \left(\frac{1}{2} - 3 \right) \right] = -\frac{3}{2} \ln 2
 \end{aligned}$$

(b) Evaluate the iterated triple integral $\int_{-2}^2 \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} \int_0^{\sqrt{x^2+y^2}} e^z dz dy dx$ by using cylindrical coordinates.

$$W = \{(x, y, z) \mid -2 \leq x \leq 2, -\sqrt{4-x^2} \leq y \leq \sqrt{4-x^2}, 0 \leq z \leq \sqrt{x^2+y^2}\}$$

$$(r, \theta, z) \quad \begin{cases} x = r \cos \theta, & y = r \sin \theta, & z = z \end{cases}$$

$$W^* = \{(r, \theta, z) \mid 0 \leq r \leq 2, 0 \leq \theta \leq 2\pi, 0 \leq z \leq r\}$$

$$\begin{aligned}
 \iiint_W e^z dz dy dx &= \int_0^2 \int_0^{2\pi} \int_0^r e^z r dz d\theta dr \\
 &= \int_0^2 \int_0^{2\pi} r [e^z]_{z=0}^{z=r} d\theta dr \\
 &= \int_0^2 \int_0^{2\pi} r(e^r - 1) d\theta dr \\
 &= 2\pi \int_0^2 r(e^r - 1) dr \\
 &= 2\pi \left[[r(e^r - r)]_{r=0}^{r=2} - \int_0^2 (e^r - r) dr \right] \\
 &= 2\pi \left(2(e^2 - 2) - [e^r - \frac{r^2}{2}]_{r=0}^{r=2} \right) \\
 &= 2\pi (2(e^2 - 2) - (e^2 - 2) + 1) = 2\pi(e^2 - 1)
 \end{aligned}$$

Problem 4. (20 points)

(a) Evaluate the scalar line integral $\int_C (x^2 + y^2 + 2z^2 + 1) ds$ where $\mathbf{x}(t) = (\cos t, \sin t, t)$ with $0 \leq t \leq 3\pi$.

$$\mathbf{x}(t) = (\cos t, \sin t, t), \quad \mathbf{x}'(t) = (-\sin t, \cos t, 1) \\ \Rightarrow \|\mathbf{x}'(t)\| = \sqrt{(-\sin t)^2 + (\cos t)^2 + 1^2} = \sqrt{2}$$

$$\begin{aligned} \int_C (x^2 + y^2 + 2z^2 + 1) ds &= \int_0^{3\pi} (\cos^2 t + \sin^2 t + 2t^2 + 1) \cdot \sqrt{2} dt \\ &= \int_0^{3\pi} \sqrt{2} (2t^2 + 2) dt \\ &= \sqrt{2} \left[\frac{2}{3} t^3 + 2t \right]_{t=0}^{t=3\pi} \\ &= 2\sqrt{2} \left(\frac{1}{3} \cdot (27\pi^3) + 3\pi \right) \\ &= 2\sqrt{2} (9\pi^3 + 3\pi) = 6\sqrt{2}\pi (3\pi^2 + 1) \end{aligned}$$

(b) Evaluate $\int_C xy dx - xy^2 dy$ where C is the curve with equation $y^3 = x^2$, from $(-1, 1)$ to $(1, 1)$.

$$\begin{aligned} \mathbf{x}(t) &= (t^3, t^2), \quad -1 \leq t \leq 1 \Rightarrow \mathbf{x}'(t) = (3t^2, 2t) \\ \int_C xy dx - xy^2 dy &= \int_{-1}^1 (t^3)(t^2)(3t^2) dt - (t^3)(t^2)(2t) dt \\ &= \int_{-1}^1 3t^7 dt - 2t^8 dt \\ &= \left[\frac{3}{8} t^8 - \frac{2}{9} t^9 \right]_{t=-1}^{t=1} \\ &= \left(\frac{3}{8} - \frac{2}{9} \right) - \left(\frac{3}{8} + \frac{2}{9} \right) \\ &= -\frac{4}{9} \end{aligned}$$

Problem 5. (20 points)

(a) Use **Green's theorem** to compute the line integral $\oint_C (y^2 - 2xy) dx + (x^2 + 2xy) dy$ where C is the boundary of the square with vertices $(0, 0)$, $(2, 0)$, $(2, 2)$, and $(0, 2)$, oriented counterclockwise.

$$\begin{aligned} & \oint_C (y^2 - 2xy) dx + (x^2 + 2xy) dy & D = \{(x, y) \mid 0 \leq x \leq 2, 0 \leq y \leq 2\} \\ & & C = \partial D \\ & = \iint_D \left(\frac{\partial}{\partial x} (x^2 + 2xy) - \frac{\partial}{\partial y} (y^2 - 2xy) \right) dx dy \\ & = \int_0^2 \int_0^2 (2x + 2y - 2y + 2x) dx dy \\ & = \int_0^2 \int_0^2 4x dx dy = \int_0^2 [2x^2]_{x=0}^{x=2} dy \\ & = \int_0^2 8 dy = 2 \cdot 8 = 16 \end{aligned}$$

(b) Given a vector field $\mathbf{F}(x, y) = (xy^2)\mathbf{i} + (x^2y + y)\mathbf{j}$. Show that $\mathbf{F}$ is a conservative vector field and then find a potential function f such that $\mathbf{F} = \nabla f$.

$$\begin{aligned} M &= xy^2, \quad N = x^2y + y \\ \frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} &= \frac{\partial}{\partial x} (x^2y + y) - \frac{\partial}{\partial y} (xy^2) = 2xy - 2xy = 0 \\ &\Rightarrow \mathbf{F} \text{ is conservative} \\ \nabla f = \mathbf{F} &\Rightarrow \begin{cases} \frac{\partial f}{\partial x} = xy^2 & \text{--- (1)} \\ \frac{\partial f}{\partial y} = x^2y + y & \text{--- (2)} \end{cases} \\ \text{from (1)} &\Rightarrow f(x, y) = \int xy^2 dx = \frac{x^2y^2}{2} + g(y) \quad \text{--- (3)} \\ \text{put (3) into (2)} &\Rightarrow \frac{\partial}{\partial y} \left(\frac{x^2y^2}{2} + g(y) \right) = x^2y + y \\ x^2y + g'(y) &= x^2y + y \\ g'(y) &= y \Rightarrow g(y) = \int y dy = \frac{y^2}{2} + C \\ \text{Thus } f(x, y) &= \frac{x^2y^2}{2} + \frac{y^2}{2} + C \end{aligned}$$